

```
DiscreteHopfieldNetwork.m X +
C:\Users\Afroz\Desktop\NN\EXP9\Code\DiscreteHopfieldNetwork.m

1  clc;
2  clear;
3  close all;
4
5  % Discrete Hopfield Network
6  % Stored Pattern = [1 1 1 0]
7
8  %% Step 1: Define the stored pattern
9  pattern = [1 1 1 0];
10
11  % Convert binary pattern to bipolar pattern
12  % 0 -> -1, 1 -> +1
13  bipolar_pattern = 2 * pattern - 1;
14
15  disp('Stored Pattern:');
16  disp(pattern);
17
18  disp('Bipolar Stored Pattern:');
19  disp(bipolar_pattern);
20
21  %% Step 2: Calculate Weight Matrix
22  W = bipolar_pattern' * bipolar_pattern;
23
24  % Set diagonal elements to zero
25  W = W - diag(diag(W));
26
27  disp('Weight Matrix:');
28  disp(W);
29
30  %% Step 3: Define input pattern with mistakes
31  % Mistakes are introduced in first and second positions
32  test_pattern = [0 0 1 0];
33
34  % Convert test pattern to bipolar form
35  test_bipolar = 2 * test_pattern - 1;
36
37  disp('Input Pattern with Mistakes:');
38  disp(test_pattern);
39
40  disp('Bipolar Input:');
41  disp(test_bipolar);
42
43  %% Step 4: Recall the stored pattern
44  x = test_bipolar';
45
46  net = W * x;
47
48  % Apply bipolar activation function
49  output = sign(net);
50
51  output(output == 0) = 1;
52
53  %% Step 5: Convert bipolar output back to binary
54  final_output = (output' + 1) / 2;
55
56  disp('Net Input:');
57  disp(net);
58
59  disp('Recovered Bipolar Pattern:');
60  disp(output);
61
62  disp('Recovered Binary Pattern:');
63  disp(final_output);
64
65  %% Step 6: Compare with stored pattern
66  if isequal(final_output, pattern)
67      disp('Stored pattern successfully recalled.');
```

## Command Window

Stored Pattern:

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 0 |
|---|---|---|---|

Bipolar Stored Pattern:

|   |   |   |    |
|---|---|---|----|
| 1 | 1 | 1 | -1 |
|---|---|---|----|

Weight Matrix:

|    |    |    |    |
|----|----|----|----|
| 0  | 1  | 1  | -1 |
| 1  | 0  | 1  | -1 |
| 1  | 1  | 0  | -1 |
| -1 | -1 | -1 | 0  |

Input Pattern with Mistakes:

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 1 | 0 |
|---|---|---|---|

Bipolar Input:

|    |    |   |    |
|----|----|---|----|
| -1 | -1 | 1 | -1 |
|----|----|---|----|

Net Input:

|   |   |    |   |
|---|---|----|---|
| 1 | 1 | -1 | 1 |
|---|---|----|---|

Recovered Bipolar Pattern:

|   |   |    |   |
|---|---|----|---|
| 1 | 1 | -1 | 1 |
|---|---|----|---|

Recovered Binary Pattern:

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 0 | 1 |
|---|---|---|---|

Pattern is not completely recalled.

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